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Research Methods in Education

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Research Title:

Enhancing Reading Proficiency Among Grade 2 Struggling Readers Through Reading Intervention Materials (RIM) in Albay Central School

METHODOLOGY

This section presents the research design, research instruments, data gathering procedures, respondents, sampling technique, study site, and data analysis plan that will be used to address the objectives of the study.

Research Design

This study will employ a quantitative approach using a pre-experimental, one-group pretest–posttest design, complemented by descriptive-evaluative and qualitative components. In a one-group pretest–posttest design, a single group is measured on the dependent variable before and after exposure to a treatment, and the change between the two measurements is examined (Campbell & Stanley, 1963; Fraenkel et al., 2012). In this study, the group is the Grade 2 struggling readers of Albay Central School, the treatment is the use of the Reading Intervention Materials (RIM), and the dependent variable is reading proficiency in letter-sound knowledge, word recognition, reading fluency, and reading comprehension. The design is represented as follows:

$$O_1 \quad X \quad O_2$$

where O_1 is the pre-test, X is the intervention using the RIM, and O_2 is the post-test. The descriptive-evaluative component will determine the level of acceptability of the RIM as rated by teachers and reading experts, while the qualitative component will document the challenges encountered by the teacher and the learners in using the materials. Embedding qualitative data within an intervention study in this way helps explain how the intervention worked under actual classroom conditions (Creswell & Creswell, 2018).

This design is the most appropriate for the study for three reasons. First, the study seeks to determine whether the reading proficiency of the same learners improves after the intervention, which requires measuring each learner before and after the use of the RIM. Second, the study is a classroom-based intervention conducted by a teacher-researcher in a single school, where the number of identified struggling readers is limited and the formation of a separate, equivalent control group is not feasible. Third, and most importantly, withholding a reading intervention from learners who have been identified as needing it, solely to create a comparison group, would be ethically questionable, since reading difficulties that are left unaddressed in the early grades tend to persist and worsen (Juel, 1988; Torgesen, 2004).

Because the design has no control group, it is vulnerable to threats to internal validity such as history, maturation, testing, instrumentation, and attrition (Campbell & Stanley, 1963). To reduce these threats, the study will (a) use two equivalent forms of the reading assessment for the pre-test and post-test to minimize practice effects; (b) administer both tests under the same standardized conditions; (c) have a second trained rater score audio-recorded oral reading without knowing whether each recording came from the pre-test or the post-test; (d) keep the intervention period short and clearly defined; (e) record attendance and all reading activities of the learners in an implementation log; and (f) report effect sizes alongside tests of significance. These safeguards will allow the results to be interpreted with appropriate confidence and caution.

Research Instrument

The study will use three researcher-made instruments, each aligned with specific objectives. The RIM itself will be described through documentary analysis of its content, activities, and mode of delivery.

Reading Proficiency Assessment (RPA). This individually administered test will measure the reading proficiency of the learners before and after the intervention. It will be prepared in two equivalent forms, Form A for the pre-test and Form B for the post-test, with parallel items of comparable difficulty aligned with the Grade 2 reading competencies of the MATATAG Curriculum (Department of Education, 2024). The test has four parts. Part I, Letter-Sound Knowledge (20 items), asks the learner to give the sound of each letter shown. Part II, Word Recognition (30 items), asks the learner to read aloud 20 decodable words and 10 high-frequency sight words presented in isolation; a word is scored correct if read accurately within three seconds. Part III, Reading Fluency, asks the learner to read aloud a Grade 2-level passage, which is audio-recorded and scored for

accuracy (percentage of words read correctly) and rate (words correct per minute), following the procedures of oral reading fluency assessment (Hasbrouck & Tindal, 2006) and the Philippine Informal Reading Inventory (Department of Education, 2018). Part IV, Reading Comprehension (10 items), asks the learner to answer literal and inferential questions orally after reading the whole passage. Parts I, II, and IV are scored one point per correct response.

Acceptability Evaluation Tool. This tool will be used by the evaluators to rate the RIM. Part I gathers the profile of the evaluator, namely position, years of teaching experience, and area of specialization. Part II contains 20 indicators distributed equally across the four criteria of content, format and design, language and readability, and usefulness, each rated on a five-point scale from 5 (Very Highly Acceptable) to 1 (Not Acceptable). Part III provides space for comments and suggestions for improving the materials.

Implementation Log and Interview Guide. To identify the challenges encountered in using the RIM, the researcher will keep an implementation log that records the date, duration, attendance, materials used, learner responses, difficulties observed, and brief reflective notes for every session. At the end of the intervention, short, child-friendly interviews will be conducted with the learners in the language they are most comfortable with, using a semi-structured guide of five to seven simple questions on which activities they found easy, difficult, or enjoyable.

Validation of the Instruments. To establish credible and unbiased data, the instruments will undergo three stages of validation. First, a panel of six experts, composed of a master teacher in reading, the school reading coordinator, a Filipino teacher, an English teacher, a school head or education program supervisor, and a research expert, will rate the relevance of each item on a four-point scale. The item-level content validity index (I-CVI) and the scale-level average (S-CVI/Ave) will be computed; items with an I-CVI of at least .78 will be retained, and an S-CVI/Ave of .90 or higher will indicate excellent content validity (Lynn, 1986; Polit & Beck, 2006). Items that fall short will be revised or replaced based on the experts' comments. Second, both forms of the RPA will be pilot-tested with 20 to 30 Grade 2 learners from a comparable class or school who are not part of the study. The internal consistency of Parts I, II, and IV will be estimated using the Kuder–Richardson Formula 20 (Kuder & Richardson, 1937), with a coefficient of .70 or higher considered acceptable, and the two forms will be considered equivalent if they yield comparable means and a strong positive correlation. The Acceptability Evaluation Tool will be pilot-tested with teachers who will not serve as evaluators, and its reliability

will be estimated using Cronbach's alpha (Cronbach, 1951). Third, to guard against scorer bias, a second trained rater, unaware of whether a recording came from the pre-test or the post-test, will independently score at least 20 percent of the oral reading recordings; an inter-rater agreement of at least 90 percent will be required, and disagreements will be resolved through discussion and re-scoring.

Data Gathering Procedures

The data gathering will proceed in six phases and is expected to last about twelve weeks within the third quarter of School Year 2026–2027.

Phase 1: Securing Permission. After the approval of the research proposal by the Graduate School of Bicol College, the researcher will submit a letter requesting permission to conduct the study to the Schools Division Superintendent, through the Public Schools District Supervisor, in accordance with the DepEd research management guidelines (Department of Education, 2017). Upon approval, the researcher will coordinate with the principal of Albay Central School on the schedule, venue, and conduct of the study.

Phase 2: Informed Consent and Ethical Safeguards. An orientation will be held for the parents or guardians of the identified learners to explain the purpose, procedures, benefits, and minimal risks of the study. Written informed consent will be obtained from the parents, and the verbal assent of each learner will be sought in simple language. Participation will be voluntary, and learners may withdraw at any time without any effect on their grades or on their access to regular reading support. In compliance with the Data Privacy Act of 2012 (Republic Act No. 10173, 2012), names will be replaced with codes, and all records and recordings will be stored securely and accessed only by the researcher and the second rater. All sessions will observe the DepEd Child Protection Policy (Department of Education, 2012). Evaluators will likewise be informed of the purpose of the evaluation and the confidentiality of their responses.

Phase 3: Validation, Pilot Testing, and Pre-test (Weeks 1–3). The instruments will be validated and pilot-tested as described above. In Week 3, Form A of the RPA will be administered individually to each learner in a quiet room, at a time that does not conflict with regular classes, following standardized directions. Each administration will take about 15 to 20 minutes. The results will establish the baseline reading proficiency of each learner and guide the grouping of learners for the intervention.

Phase 4: Implementation of the Intervention (Weeks 4–11). The RIM will be used for eight weeks in sessions of about 30 minutes held four times a week, for a total of

about 32 sessions. Sessions will be scheduled, in coordination with the school head, during periods such as remedial reading time and Catch-Up Fridays so that learners do not miss regular instruction. Learners will work in small groups of three to five formed according to their pre-test results, with one-on-one sessions for those who need closer guidance. Each session will follow a consistent routine of review, explicit teaching of a new skill, guided practice, independent practice, and a brief check of learning. Attendance, activities, learner responses, and difficulties will be recorded in the implementation log after every session.

Phase 5: Post-test, Evaluation, and Interviews (Week 12). Form B of the RPA will be administered under the same conditions as the pre-test. The RIM and the Acceptability Evaluation Tool will be distributed personally to the evaluators, who will be given one week to examine the materials and accomplish the tool independently. The learner interviews will also be conducted during this week.

Phase 6: Retrieval, Extraction, and Analysis of Data. All accomplished instruments will be retrieved personally by the researcher and checked for completeness. Scores and ratings will be encoded in a spreadsheet and verified through double entry, while log entries and interview responses will be transcribed and organized for thematic analysis. The data will be analyzed using the tools described in the data analysis plan, kept confidential, used solely for this study, and properly disposed of after its completion. The results will be shared with the school and the parents, and learners who still need help will continue to receive reading support after the study.

Respondents of the Study

The study has two groups of respondents. The first group, the subjects of the intervention, consists of the Grade 2 struggling readers of Albay Central School in School Year 2026–2027. They will be identified through the beginning-of-the-year reading assessment administered by the school and confirmed through the pre-test. The second group consists of the evaluators of the RIM, composed of teachers and reading experts who will rate the acceptability of the materials.

Learners will be included if they (a) are officially enrolled in Grade 2 at Albay Central School in School Year 2026–2027; (b) were identified as reading below grade level in the beginning-of-the-year reading assessment; (c) have the written consent of their parents or guardians and give their own assent; and (d) are expected to remain enrolled for the duration of the study. Learners will be excluded if they (a) have a diagnosed condition, such as an uncorrected visual or hearing impairment or a learning

disability, that requires specialized intervention beyond the scope of the RIM, in which case they will be referred for appropriate support; (b) transfer out of the school during the study; or (c) attend fewer than 80 percent of the intervention sessions, in which case their data will be excluded from the pre-test and post-test comparison, although they will continue to receive reading support.

Ten evaluators will be selected based on the following criteria: (a) current assignment as a Grade 1 to Grade 3 teacher, master teacher, reading coordinator, or school head; (b) at least five years of experience in teaching reading in the early grades or in supervising reading programs; and (c) no involvement in the development of the RIM, to ensure an objective evaluation.

Sampling Technique

The study will use purposive sampling, a non-probability technique in which participants are deliberately selected because they possess characteristics that are relevant to the purpose of the study (Etikan et al., 2016). Since the study aims to enhance the reading proficiency of Grade 2 learners who are struggling in reading, only learners who meet the inclusion criteria will qualify. To ensure that the sample represents the whole target population, the researcher will use complete enumeration: all Grade 2 learners of Albay Central School who meet the criteria will be included, rather than only a portion of them. The sample is therefore the entire population of identified Grade 2 struggling readers in the school, which eliminates sampling error within the school and strengthens the credibility of the findings. The exact number of learners will be determined after screening and the securing of consent.

The evaluators will likewise be chosen through purposive sampling, since the credibility of the acceptability ratings depends on their expertise in early reading instruction. Because all respondents come from a single school, the findings will be generalized only to the Grade 2 struggling readers of Albay Central School, although they may serve as a reference for schools with similar contexts.

Study Site

The study will be conducted at Albay Central School, a public elementary school under the Department of Education in the Province of Albay, Bicol Region. Several features make the school an interesting and appropriate site for the study. First, it serves young learners who mostly speak Bikol at home but are learning to read in Filipino and English, which provides a meaningful setting for testing contextualized reading materials. Second, the school’s beginning-of-the-year reading assessment has identified Grade 2

learners who are reading below grade level, establishing a clear and immediate need for intervention. Third, the school implements DepEd reading initiatives such as Catch-Up Fridays and reading remediation, which provide built-in time and structure into which the RIM can be integrated. Finally, the school is the researcher's own workplace, which allows daily access to the learners, close coordination with the school head, teachers, and parents, and consistent implementation of the intervention throughout the study period.

The choice of the site is consistent with the DepEd research management guidelines, which encourage teachers to conduct research that informs and improves teaching and learning in their own schools (Department of Education, 2017), and with the nature of classroom-based intervention research, in which the teacher-researcher studies the very learners the intervention is meant to serve.

Data Analysis Plan

The data will be analyzed according to the objectives of the study. Quantitative data will be encoded in Microsoft Excel and processed using statistical software such as SPSS or jamovi, with the level of significance set at .05.

Objectives 1 and 3. Frequency, percentage, mean, and standard deviation will be used to describe the reading proficiency of the learners in each of the four areas before and after the intervention. Scores in letter-sound knowledge and word recognition will be converted to percentages and interpreted using Table 1, while reading fluency (word reading accuracy) and reading comprehension will be interpreted using the Phil-IRI reading levels in Table 2 (Department of Education, 2018). Reading rate, in words correct per minute, will be reported descriptively.

Table 1. *Scale for Interpreting Letter-Sound Knowledge and Word Recognition Scores*

Percentage Score	Proficiency Level	Description
90–100	Proficient	Demonstrates the skill accurately and consistently
75–89	Approaching Proficiency	Demonstrates the skill with minor errors
50–74	Developing	Demonstrates the skill inconsistently; needs guidance
Below 50	Beginning	Has not yet acquired the skill; needs intensive support

Table 2. *Phil-IRI Reading Levels for Word Reading Accuracy and Comprehension*

Reading Level	Word Reading Accuracy	Comprehension
Independent	97–100%	80–100%
Instructional	90–96%	59–79%
Frustration	89% and below	58% and below

Objective 2. The RIM will be described through documentary analysis, and its content, activities, and mode of delivery will be summarized in a matrix.

Objective 4. The normality of the differences between the pre-test and post-test scores will first be examined using the Shapiro–Wilk test (Shapiro & Wilk, 1965), which is suited to small samples. If the differences are normally distributed, the paired-samples t-test will be used; otherwise, the Wilcoxon signed-rank test (Wilcoxon, 1945) will be applied. The analysis will be done for each of the four reading areas and for the overall score, and the null hypothesis of no significant difference will be rejected if $p < .05$. To determine the magnitude of improvement, Cohen's d (for the t-test) or r (for the Wilcoxon test) will be computed and interpreted using Cohen's (1988) benchmarks of small, medium, and large effects ($d = 0.20, 0.50, 0.80$; $r = .10, .30, .50$).

Objective 5. The weighted mean and standard deviation of the evaluators' ratings will be computed for each indicator, each criterion, and the overall acceptability of the RIM, and interpreted using Table 3.

Table 3. *Scale for Interpreting the Acceptability of the RIM*

Scale	Mean Range	Verbal Interpretation
5	4.21–5.00	Very Highly Acceptable
4	3.41–4.20	Highly Acceptable
3	2.61–3.40	Moderately Acceptable
2	1.81–2.60	Slightly Acceptable
1	1.00–1.80	Not Acceptable

Objective 6. Implementation log entries and interview responses will be analyzed through thematic analysis following the six phases of Braun and Clarke (2006): familiarization with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report. Frequency counts will be used to rank the challenges from the most to the least common.

Objective 7. The reading enhancement action plan will be formulated by synthesizing the results of Objectives 1 to 6, matching the identified needs and challenges

with specific strategies, persons responsible, time frames, resources, and success indicators.

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